Rodolfo Ostilla Mónico

Associate Professor (Profesor Titular), Fluid Mechanics

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Summary

Fluid dynamicist specializing in the direct numerical simulation of turbulent flows.

Education

Doctor of Philosophy (PhD) — **Twente University**, Physics of Fluids — Enschede, NL, 2011 – 2015
- Thesis: Highly turbulent Taylor-Couette flow: direct numerical simulations
- Advisors: D. Lohse (director), R. Verzicco (co-director)

Aerospace Dynamics MSc — **Cranfield University**, Aerodynamics — Cranfield, UK, 2009 – 2010
- Thesis: Aerodynamic flow control through weak plasma interaction

Ingeniero Aeronáutico (5-year degree) — **Universidad de Sevilla**, Aerospace Engineer — Sevilla, Spain, 2005 – 2010

Grado en Filosofía (Philosophy BSc) — **Universidad Nacional de Educación a Distancia (UNED)**, Philosophy — Spain, 2015 – 2022

Experience

Universidad de Cádiz, Assistant Professor / Profesor Titular — Puerto Real, Spain, 2026 – present, 4 months
- Department of Mechanical Engineering and Industrial Design, School of Engineering

Universidad de Cádiz, Emergia Postdoctoral Fellow — Puerto Real, Spain, 2022 – 2026, 4 years 5 months
- Department of Mechanical Engineering and Industrial Design, School of Engineering

Dept. of Mechanical Engineering, University of Houston, Assistant Professor — Houston, USA, 2017 – 2022, 4 years 5 months
- Taught undergraduate and graduate courses in fluid mechanics and non-linear dynamics
- Supervised 5 MSc theses, 4 undergraduate research projects, and 1 PhD thesis

Harvard University, Postdoctoral Researcher — Cambridge, MA, USA, 2015 – 2017, 2 years

European Space Astronomy Centre (ESAC), Trainee — Madrid, Spain, 2010 – 2011, 6 months

Departamento de Mecánica de Fluidos, Universidad de Sevilla, Research Assistant — Seville, Spain, 2008 – 2009, 1 year 1 month

Projects

Aeroacústica e hidrodinámica de los instrumentos musicales tipo chimenea — 2023 – 2027

University of Cádiz, competitive research project funded by the Spanish State Research Agency (Proyectos de Generación de Conocimiento 2022). Main PI.
- Total funding: €81,250

CDS&E: Multi-scale Coherent Structure Extraction and Tracking for Modern CFD Data Analysis 2021 – 2024
University of Houston, competitive research project funded by the U.S. National Science Foundation. Co-PI with Guoning Chen and Di Yang.
- Total funding: $453,000

Particulate Rayleigh-Bénard Flow 2021 – 2024
University of Houston, competitive research project funded by the U.S. National Science Foundation. Principal Investigator.
- Total funding: $209,666

Controlling Secondary Flows by the Use of Non-Wetting Surfaces 2019 – 2022
University of Houston, competitive research project funded by the U.S. National Science Foundation. Co-PI with Kamran Alba.
- Total funding: $340,000

Awards

Visions of Fluid Dynamics award (European Fluid Dynamics Conference) 2025
With W. Shen and X. Zhu, for the video “Vortices vs. Magnetic Fields: Competing Orders in Flux Tubes”.

Emergia Fellowship 2022
Fellowship funding return to Spain and research at the Universidad de Cádiz (€85,000).

College Junior Teaching Award 2021

College Junior Research Award 2021

Vizapalooza competition winner 2018

Gallery of Fluid Motion award 2017
For the video “The Emergence of Small Scales in Vortex Ring Collisions”.

ERCOFTAC Da Vinci Award finalist 2015

Wim Nieuwpoort Award for best national scientific code 2012

Best MSc thesis (departmental prize) 2010

Royal Aeronautical Society Aerospace Challenge, winning team 2010

Teaching

- Universidad de Cádiz: Mecánica de Fluidos II (Grado en Ingeniería Aeroespacial, 4.5 credits), 2022/23 to 2025/26. Course redesigned by me; covers potential flow, boundary layers, compressible flows, turbulence and CFD labs.
- Universidad de Cádiz: Instalaciones y Máquinas Hidráulicas y Térmicas (Máster Universitario en Ingeniería Industrial, 5 ECTS), hydraulic part, 2026/27 (current).
- Universidad de Cádiz: short course "An Invitation to Computational Fluid Dynamics from Fundamentals to Applications" (20 hours), September 2026.
- University of Houston: Introduction to Fluid Mechanics (2018 to 2021), Fluid Mechanics II / Fluids II (2018, 2019), and Advanced CFD (2019, graduate course developed by me).
- University of Houston: Scientific Computing and Non-linear Dynamics and Chaos (combined undergraduate/graduate, 2020).

Supervision

- Doctoral thesis: V. Jeganathan, “Controlling Secondary Flows in Turbulent Taylor-Couette Flow Using Surface Heterogeneity” (University of Houston, 2022).
- Master theses (University of Houston): S. Soriano (2022), D. Putt (2021), A. Janjua (2020), H. Mitra (2020), A. Mishra (2019).
- Undergraduate theses (TFG), Universidad de Cádiz: main director of 12 completed projects (2025 to 2026) on ventilation, vortex dynamics, CFD of rotors, wings and tsunami flows, and indoor air quality; co-director of 2 more.
- Undergraduate research fellows (University of Houston): N. Betancourt, S. Soriano, R. Bhattarai, C. Tolat.

Service

- Chair (2025), Vice-Chair, and member (2022 to 2025) of the Diversity and Inclusion Committee, APS Division of Fluid Dynamics.
- Session organizer and chair at APS-DFD, Euromech and Spanish Fluid Mechanics Conference meetings; scientific committee of the 22nd International Couette-Taylor Workshop (2023).
- Reviewer for over 100 papers in leading fluid mechanics and physics journals; proposal reviewer for the NSF (US), the European Research Council and Fondecyt (Chile); judge of the Gallery of Fluid Motion (2018, 2021).
- Faculty advisor of the aeromodelling student association UCa&Air, Universidad de Cádiz.

Publications

Elastic turbulence in space
Rodolfo Ostilla-Mónico
doi.org/10.1017/jfm.2024.1214

Fluid dynamical pathways of airborne transmission while waiting in a line
Ruixi Lou, Milo Van Mooy, Gabriel A. Tarditti, Rodolfo Ostilla-Mónico, Varghese Mathai
doi.org/10.1126/sciadv.adw0985

Transition between boundary-limited scaling and mixing-length scaling of turbulent transport in internally heated convection
Sina Kazemi, Rodolfo Ostilla-Mónico, David Goluskin
doi.org/10.1103/PhysRevLett.129.024501

Controlling secondary flows in Taylor-Couette flow using axially spaced superhydrophobic surfaces
Vignesh Jeganathan, Tala Shannak, Kamran Alba, Rodolfo Ostilla-Mónico
doi.org/10.1017/jfm.2023.606

Instability and disintegration of vortex rings during head-on collisions and wall interactions
Aakash Mishra, Alain Pumir, Rodolfo Ostilla-Mónico
doi.org/10.1103/PhysRevFluids.6.104702

Turbulence generation through an iterative cascade of the elliptical instability
Ryan McKeown, Rodolfo Ostilla-Mónico, Alain Pumir, Michael P. Brenner, Shmuel M. Rubinstein
doi.org/10.1126/sciadv.aaz2717

Physical and geometric constraints shape the labyrinth-like nasal cavity
David Zwicker, Rodolfo Ostilla-Mónico, Daniel E. Lieberman, Michael P. Brenner
doi.org/10.1073/pnas.1714795115

Logarithmic mean temperature profiles and their connection to plume emissions in turbulent Rayleigh-Bénard convection
Erwin P. van der Poel, Rodolfo Ostilla-Mónico, Roberto Verzicco, Siegfried Grossmann, Detlef Lohse
doi.org/10.1103/PhysRevLett.115.154501

A pencil distributed finite difference code for strongly turbulent wall-bounded flows
Erwin P. van der Poel, Rodolfo Ostilla-Mónico, John Donners, Roberto Verzicco
doi.org/10.1016/j.compfluid.2015.04.007

Optimal Taylor-Couette flow: direct numerical simulations
Rodolfo Ostilla, Richard J. A. M. Stevens, Siegfried Grossmann, Roberto Verzicco, Detlef Lohse
doi.org/10.1017/jfm.2012.596

Skills

Computational Fluid Dynamics

Applied Fluid Mechanics

Interests

Philosophy of Science

Languages

Spanish
Native speaker

English
Fluent

Mandarin Chinese
B1